Evaluation of Satellite Rainfall Products for Monsoon Extremes in the Bangladesh Portion of the Ganges-Brahmaputra-Meghna River Basin

Abstract

This project will evaluate CHIRPS and IMERG rainfall estimates against Bangladesh Meteorological Department (BMD) daily observations in the Bangladesh portion of the Meghna River basin. The analysis will focus on June–September monsoon rainfall over a proposed 2001–2023 period, subject to station coverage. A reproducible Python workflow will match stations with satellite grid cells, screen missing observations, calculate seasonal extreme-rainfall indices, and compare rainfall errors and heavy-event detection. The study will test whether similar seasonal totals conceal differences in the representation of extreme rainfall. Expected outputs include station-level performance summaries, diagnostic figures, and an assessment of which product is more suitable for particular rainfall measures. Published figures from Khairul et al. (2018) provide context for the proposed analysis.

Background and significance

Rainfall information is essential for assessing waterlogging and flood exposure in the Meghna basin, where transboundary data limitations complicate hydrological applications. Khairul et al. (2018) evaluated four gauge-adjusted products against observations in the Bangladesh portion of the basin and demonstrated the importance of evaluating and correcting rainfall errors. Montes et al. (2021) examined monsoon extremes across Bangladesh and found that product performance depends on the rainfall index. Their national evaluation motivates a focused examination of stations within the Meghna basin.

CHIRPS combines infrared satellite information and station observations (Funk et al., 2015), whereas IMERG combines multiple satellite retrievals with gauge adjustment in its Final Run. Differences between precipitation products also affect inferred changes across the GBM region (Khatiwada & Curtis, 2021). This project will provide a manageable basin-specific comparison of two products. Its contribution is the consistent evaluation of seasonal amounts, extreme-event magnitude, and daily event detection within one basin; it does not claim to be the first satellite evaluation there.

Research questions and hypothesis

1. How accurately do CHIRPS and IMERG reproduce monsoon rainfall totals and extreme-rainfall indices at eligible BMD stations?
2. How does their ability to detect heavy-rainfall days vary among stations, and does the better-performing product depend on the metric?

I hypothesize that agreement in seasonal rainfall totals will not necessarily imply agreement in extreme rainfall or event timing. Guided by Montes et al. (2021), I expect IMERG to detect heavy-rainfall days more effectively at many stations, but will test this rather than assume a basin-wide ranking.

Data and planned analysis

I will use CHIRPS v2.0 Final daily rainfall at 0.05° resolution and IMERG V07 Final Run at 0.1°. Keeping CHIRPS v2.0 makes the comparison consistent with the earlier literature. Satellite files will be downloaded for the study domain and reduced to station time series. Native grid resolutions will be retained; finer resolution will not be treated as evidence of greater accuracy. Dataset versions, download dates, units, and extraction coordinates will be recorded.

Preprocessing in Python. pandas will standardize dates and missing-value codes; GeoPandas will select stations; and xarray will extract the nearest satellite grid cell. I will check duplicate dates, negative rainfall, coordinate errors, unit conversions, and grid-cell distance. BMD accumulation times will be compared with satellite daily windows. Where necessary, half-hourly IMERG will be aggregated to the BMD day; any unresolved CHIRPS timing mismatch will be reported and examined with a one-day-shift sensitivity check.

Quality control. Missing observations will never be converted to zero. Daily comparisons will use dates valid in all three datasets. A station-year will be retained for seasonal totals and indices only when all 122 JJAS days are available in all datasets; no extreme values will be interpolated. I will aim for at least ten complete seasons per station and report all exclusions. If too few stations meet this rule, the project will use a shorter common interval and clearly limit its conclusions.

Answering Question 1. Following the index framework of Zhang et al. (2011) and its monsoon adaptation by Montes et al. (2021), I will compute total JJAS rainfall; R95p and R99p, the rainfall sums on days exceeding wet-day 95th and 99th percentiles; R95pTOT and R99pTOT, those sums divided by seasonal rainfall and multiplied by 100; and R20mm, the number of JJAS days with rainfall ≥20 mm. Wet days have rainfall ≥1 mm. Fixed station-specific thresholds will be estimated from pooled BMD wet days over the common retained period and applied to both products. Product-specific thresholds will be checked separately because they answer a different question about relative climatological extremes.

For each station, I will calculate mean bias, root-mean-square error (RMSE), and Pearson correlation for paired daily values and annual index series. Bias is mean(satellite − gauge); RMSE is the square root of mean squared differences. Relative bias is 100 × sum(satellite − gauge)/sum(gauge), with zero denominators flagged. These diagnostics follow the evaluation approaches of Khairul et al. (2018) and Montes et al. (2021). Figure 1 provides published evidence of daily rainfall errors that the proposed diagnostics will examine.

Answering Question 2. At the common 20 mm/day threshold, a hit occurs when both datasets exceed or equal the threshold, a miss when only the gauge does, and a false alarm when only the satellite does. I will calculate probability of detection = hits/(hits + misses), false-alarm ratio = false alarms/(hits + false alarms), and critical success index = hits/(hits + misses + false alarms). Undefined denominators will be flagged. Station-wise differences between products will be assessed with 1,000 paired bootstrap resamples of whole monsoon years, preserving the daily sequences and common years across stations. Wide intervals will be interpreted as uncertainty, particularly for rare R99p events.

Published evidence and event evaluation

Figures 1 and 2 present published Meghna basin results from Khairul et al. (2018). The daily scatterplots show substantial disagreement between satellite and gauge rainfall, while the monthly plots show seasonal variation and differences among products. These findings motivate separate evaluation of rainfall amount and daily extremes. The published comparisons use BWDB gauges and four products; the proposed study will use BMD observations and CHIRPS–IMERG.

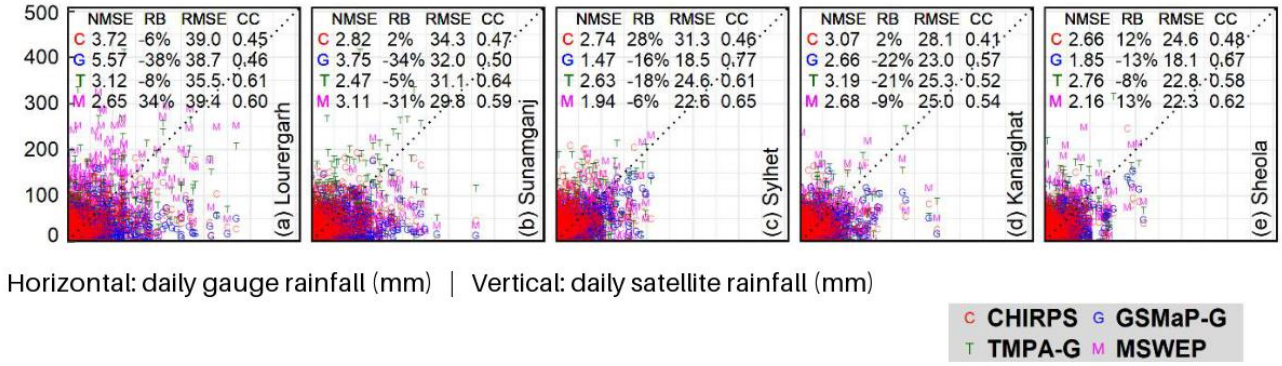


Figure 1. Daily satellite rainfall versus gauge rainfall at Lourerghat, Sunamganj, Sylhet, Kanaighat, and Sheola during the wet seasons of 2009–2016. Reproduced excerpt from Figure 5 of Khairul et al. (2018), showing its first row and original legend. Products are CHIRPS, GSMaP-G, TMPA-G, and MSWEP. Source: doi:10.3390/rs10060828 (CC BY 4.0).

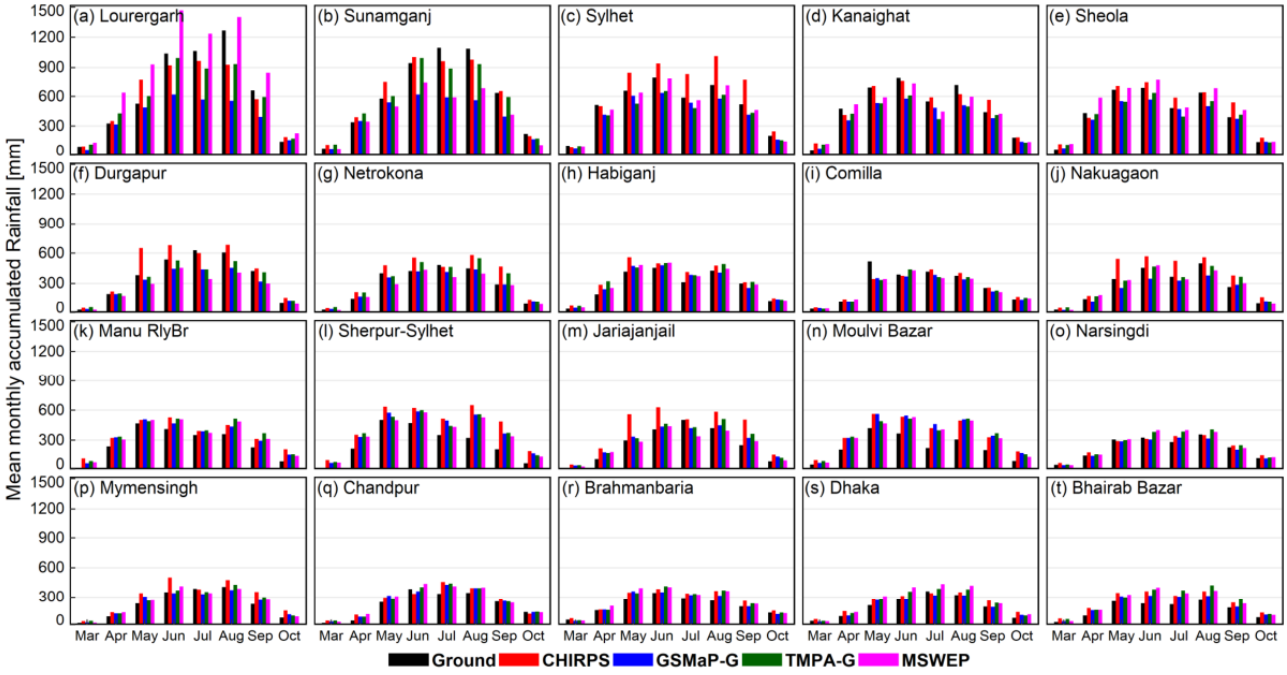


Figure 2. Mean monthly rainfall for March–October, 2009–2016, comparing ground gauges with CHIRPS, GSMaP-G, TMPA-G, and MSWEP at 20 stations. Reproduced from Figure 7 of Khairul et al. (2018), doi:10.3390/rs10060828 (CC BY 4.0). These are published results, not findings from the proposed study.

Completion timeline and feasibility

I will allocate 10–12 hours per week to programming and analysis over four weeks, with additional writing time. In week 1, I will obtain daily BMD records, verify the basin boundary and station metadata, and test downloads for one station-year. In week 2, I will extract all eligible station series, implement quality control, and verify index functions with small hand-calculated examples. In week 3, I will calculate performance metrics, bootstrap uncertainty, and generate original figures from the BMD–satellite comparisons. In week 4, I will complete the timing-sensitivity checks, interpret the results, and finalize the notebook, report, and presentation. If acquisition is delayed, I will shorten the common period rather than expand to other basins or products.

Expected outcomes and broader impacts

The project will deliver a reproducible Jupyter notebook, a documented station-selection map, paired rainfall tables, and product comparisons by station and metric. These outputs can support rainfall-product selection for local hazard assessment and subsequent hydrological studies. They will not directly validate flood forecasts or rainfall outside Bangladesh. Both products incorporate gauge information, so overlap with BMD observations may compromise independence; this will be disclosed. Gauge-to-grid scale differences, sparse station coverage, daily timing, and rare-event sampling will also limit interpretation.

References

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